Defining Data Integrity

Ensuring data integrity is paramount to developing accurate health information and cannot be overemphasized. The integrity of the data generated is directly related to what is captured, how it is captured, and the source the data came from. There are many sources of data, ranging from the patient themselves to the multitude of clinical and administrative computer systems in which data is collected. Regardless of the source, there should be consistency in how and when data is captured to ensure its integrity. Data integrity impacts the quality of care delivered to patients, so healthcare data must always be complete, accurate, consistent, and timely to achieve optimal patientcare outcomes.

Databases and Data Analytics

The performance of data analytics requires access to aggregate data or data gathered from multiple instances of an event or circumstance that is then added together. Data can be aggregated across multiple instances for a single individual, single instances from multiple individuals or multiple instances from multiple individuals.

In healthcare, data is abstracted from patient health records and aggregated to create information about patients or groups that can be evaluated or analyzed. The data is collected in one or more databases. Databases are collections of locally related data, text, references, or pictures that are organized in a standardized format for use with multiple applications.

Accurate up to date patient healthcare data assists providers in making decisions and recommendations for patient care and treatment.

Consider the following example: *Mr. Singh was involved in a motor vehicle accident and sustained a head injury. Dr. Francis wanted to order an MRI to look for internal bleeding. Fortunately, Mr. Singh's health record indicates he has a pacemaker, which means an MRI is contradicted. The integrity of the information in Mr. Singh's health record alerted Dr. Francis to the health risk of this diagnostic test.*

Ways in which databases can assist organizations include:

- Facilitate data sharing
- Streamline workflow
- Assist in clinical decision making
- Provide information for managerial decision making
- Provide data for data analysis

Data analytics examines raw data in a way to draw conclusions about the data that can be used to guide decision making. The area of data analytics uses data mining, statistics, data models and machine learning to examine data. Data analytics can be descriptive, diagnostic, predictive, or prescriptive.

Learn more about each type of analytics listed below:

Descriptive Analytics

Descriptive analytics uses statistical measurements such as the mean (average) or frequency distribution to describe the characteristics of a group of patients who have something, such as a disease, in common.

Diagnostic Analytics

Data mining or dashboards are examples of diagnostic analytic tools. Their purpose is to diagnose or determine the reason something is happening, such as why the organization has had an increase in hospital acquired conditions like infections or acute blood loss anemia.

Predictive Analytics

Predictive analytics uses historical data to predict the likelihood that a specific set of circumstances will occur. For example, in areas where hurricanes are common, predictive analytics could help an organization know what types of injuries and how many of each type to prepare for in the event of another hurricane.

Prescriptive Analytics

Prescriptive analytics uses information from descriptive and predictive analytics and data modeling to determine the best course of action or solution to a problem given a specific set of circumstances. For example, prescriptive analytics takes data gathered from disasters or health emergencies and uses it to understand how to better prepare for and mitigate the effects of the next disaster.

Consider the following example: *General Hospital wants to know if giving tissue plasminogen activator (tPA) to suspected stroke patients, upon arrival in the emergency department, is correlated with the patient having fewer residual symptoms upon discharge. Looking at the health record for a single patient may not tell the whole story regarding the effectiveness of administering tPA. However, through data analysis, the number of residual symptoms documented upon discharge can be pulled from the health records for a group of stroke patients and used to create a graph that can be compared to the results of groups who did not receive tPA upon arrival in the emergency department or compared to a national standard. The result of the comparison can then be used to determine if giving tPA to suspected stroke patients, upon arrival in the emergency department, has a positive impact on the patient's overall recovery.*

Data Retention Schedules

Health information professionals manage the retention of data in all forms of media, including paper, images, optical disk, microfilm, DVD, and CD-ROM. The sources from which data is retrieved, stored, and maintained include, but are not limited to, application-specific databases, diagnostic biomedical devices, master patient indexes, and patient medical records and health information. To ensure the availability of timely, relevant data and information for patient care purposes; to meet federal, state, and local legal requirements; and to reduce the risk of legal discovery, organizations must establish appropriate retention schedules.

At a minimum, record retention schedules must:

- Ensure patient health information is available to meet the needs of continued patient care, legal requirements, research, education, and other legitimate uses of the organization
- Include guidelines that specify what information is kept, the period for which it is kept, and the storage medium on which it will be maintained (e.g., paper, microfilm, optical disk, magnetic tape)
- Include clear destruction policies and procedures that include appropriate methods of destruction for each medium on which information is maintained

Federal, state, and accrediting agencies' retention requirements will influence the determination of retention timeframes. Since there are no across-the board retention standards, all relevant requirements must be reviewed to create a compliant retention program. Health record retention schedules should require retention of records according to the timeframe of the most restrictive requirement. For example, if an accrediting agency's retention requirement for a health record is 7 years, but the retention requirement for the state where the organization is based is 10 years, the organization must retain the record for at least 10 years, if no more restrictive requirement (such as retention of records for minor patients) is in place.

Special patient populations such as minors, behavioral health or research study patients may have their own retention requirements that must be followed aside from regular retention schedules.

Retention and Destruction of Health Information

"Health information management professionals traditionally have performed retention and destruction functions using all media, including paper, images, optical disk, microfilm, DVD, and CD-ROM. The warehouses or resources from which to retrieve, store, and maintain data and information include, but are not limited to, application-specific databases, diagnostic biomedical devices, master patient indexes, and patient medical records and health information. To ensure the availability of timely, relevant data and information for patient care purposes; to meet federal, state, and local legal requirements; and to reduce the risk of legal discovery, organizations must establish appropriate retention and destruction schedules."

AHIMA Model E-Discovery Policies

"The purpose of this policy is to achieve a complete and accurate accounting of all relevant records within the organization; to establish the conditions and time periods for which paper-based and electronic health information and records will be stored, retained, and destroyed after they are no longer active for patient care or business purposes; and to ensure appropriate availability of inactive records.

This policy applies to all enterprise health information and records whether the information is paper-based or electronic. It applies to any health record, regardless of whether it is maintained in the Health Information Management Department or by the clinical or ancillary department that created it.

It is the policy of this organization to maintain and retain enterprise health information and records in compliance with applicable governmental and regulatory requirements. The organization will adhere to

retention schedules and destruction procedures in compliance with regulatory, business, and legal requirements."

Data Governance

"Healthcare organizations increasingly struggle to manage huge volumes of e-mail, hundreds of thousands of documents on shared drives, and hundreds of source and feeder systems that pour data and records into EHR systems, data warehouses, and decision support systems. This has given new panache to the old idea that governing data and data management should be the cornerstone of information services management.

Data governance is the set of policies and procedures that determine the who, how, and why of data management within the organization. Strong data governance supports compliance and legal efforts by organizing data for retrieval and retention, especially over the long term."

Data Quality and Integrity

Every day in the United States, patients' health data is collected, shared, and used in different ways throughout the healthcare continuum. The health information (HI) professional plays a vital role in managing the vast amount of clinical, financial, or administrative data that is collected and the information derived from this data. The electronic health record (EHR) is the primary repository of the data that is collected over time, which is used to build a patient's longitudinal health record and provide the basis for care that is appropriate for each specific patient. As guardians of the legal health record, HI professionals should focus on maintaining the integrity and quality of the health data collected and safeguard all information in the health record.

Master Patient Index

When patients present to a hospital or other place of healthcare delivery for healthcare services, their health record must be located or newly created. The master patient index (MPI), which has evolved over time from a classic rolodex to sophisticated computer software, is the means of locating a patient's unique health data. The integrity of the MPI is crucial to providing appropriate patient care.